

Classification and Repetition Counting in Rehabilitative Shoulder Exercises Using Smartphone Sensors

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Abstract

This project investigates the feasibility of using the smartphone, a ubiquitous device, secured with an arm-band to monitor and recognize three key shoulder rehabilitative exercises: external rotation stretches, "window washing", and scaption raises. By using integrated accelerometer and gyroscope sensors inside users' smartphones, we developed a Machine Learning (ML) pipeline using the Random Forest Classifier model, and a peak counting algorithm to classify the users' movements into one of the three aforementioned exercises, as well as count the number of repetitions performed. Our results demonstrate a high degree of accuracy in activity recognitions, ultimately achieving a 91% classification accuracy across offline participant test data. While the peak-detection algorithm was designed to count repetitions, we observed notable inconsistencies that may be attributed to smartphone hardware. Overall, the findings suggest that ubiquitous computing devices (i.e., the smartphone) may be a pathway towards at-home rehabilitation, and a bridge between clinical therapy and individual recovery.

1 Introduction

Shoulder pain is one of the most common musculoskeletal complaints [17], and therapeutic exercise is frequently prescribed as treatment. However, patients may struggle with unsupervised home rehabilitation due to a lack of feedback and inconsistent form. In this paper, we investigated how a smartphone on an arm band could be used to recognize common shoulder rehabilitation exercises using accelerometer and gyroscope data. Using a ubiquitous device, such as the smartphone, we aimed to develop a more accessible rehabilitative experience for users outside of the clinical setting. We focused on three specific exercises: an external rotation stretch, a "window washing" exercise, and scaption raises.

The main objective of this paper was to understand how the collection of sensor data can be used to identify the arm exercise being performed and keep count of the number of repetitions. We aimed to develop a Machine Learning (ML) pipeline that could increment a counter when a movement was performed, providing users with

feedback on their session. In order to do this, the team performed these exercises while having a phone attached to each user's arm. Each of these exercises involved a different type of arm movement in order to gather a wide range of accelerometer and gyroscope data. Our technical approach followed the Activity Recognition Chain (ARC) [8] to filter out noise and segment sensor data before feeding it into the pipeline for identification and classification.

2 Background

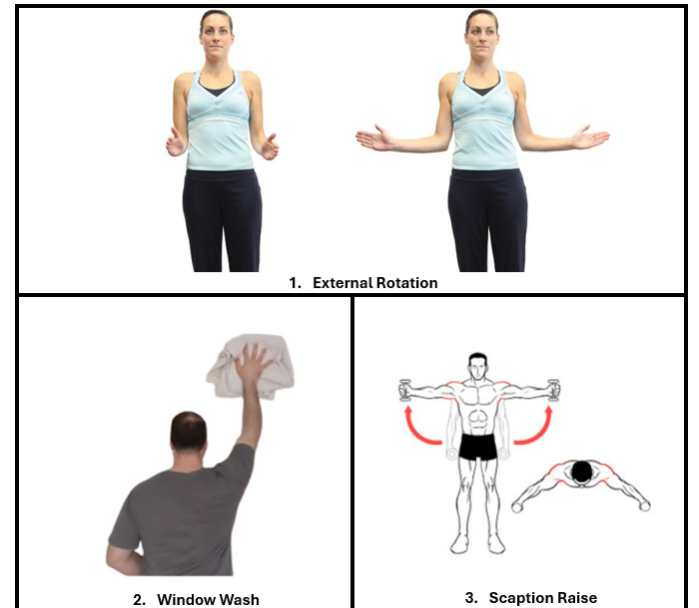


Figure 1: The shoulder rehabilitation exercises that will be analyzed in this project. Images adapted from [2] [7] [3].

Shoulder pain is estimated to be the third most common musculoskeletal complaint behind low back and knee pain [17]. Therapeutic exercise is often used by physicians as a first line of treatment in the management of shoulder pain [18]. Shoulder rehabilitation exercises focus on keeping the shoulder joint stable by strengthening supporting muscles, restoring range of motion, and preventing further injury [6].

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Modern smartphones come equipped with inertial sensors such as accelerometers and gyroscopes. Data from these sensors can be leveraged to identify and classify different types of arm movements during exercise [19]. A common approach for smartphone-based monitoring of exercise is to secure the phone to the upper arm using an arm band.

By combining these observations (rehabilitation exercises, smartphone sensors, and arm band placement) we arrived at the core idea for this project. That is, a smartphone anchored to the upper arm can capture arm movements during shoulder exercises, enabling classification of exercise type and counting of repetitions. This approach could address challenges like inconsistent exercise form, incomplete adherence to home exercise programs, and lack of feedback for patients performing unsupervised rehabilitation [9].

For this project, we focused on collecting and analyzing data for three common shoulder rehabilitation exercises, shown in Figure 1:

- (1) An external rotation stretch, intended to restore range of motion and prevent injury [5].
- (2) A "window washing" exercise, aimed at restoring shoulder range of motion and improving upward rotation of the scapula [7].
- (3) An arm raise performed in the scapular plane (hence "scaption"), intended to develop strength and stability in the shoulders and shoulder blades [10].

2.1 Relevant Literature

Muehlbauer et al. [19] showed that a single smartphone worn on the upper arm could be used to recognize and monitor arm-based exercises. Their methodology of segmenting continuous sensor data to isolate exercise periods, extracting time- and frequency-domain features, and applying machine learning classification (e.g., k-NN) combined with peak-based repetition counting showed high accuracy at correctly recognizing which exercises were being performed (over 93%) and could estimate repetition counts relatively well, with 92% falling within a 20% deviation. This shows us that effective shoulder exercise monitoring does not need multiple body-mounted sensors and that using only a smartphone is feasible.

DeVrio et al. [11] demonstrated that using a smartphone and smartwatch in conjunction can accurately track arm joint positions without the use of more specialized equipment. The authors list rehabilitation as a target application, and describe how continuous tracking could be used to monitor metrics like total motion, maximum extension, and maximum height reached. While our work relies on a single smartphone rather than a smartphone/smartwatch pair, SmartPoser's rehabilitation use case and arm motion metrics motivate our approach of using smartphone sensor data to classify shoulder rehab exercises and count repetitions.

Ongevise et al. [20] developed a smartphone-based system for patients with frozen shoulder which demonstrated that a phone strapped to the elbow can effectively track shoulder range of motion during exercise. Using only an accelerometer, they were able to estimate arm rotation angle with high accuracy. They also created a web interface for physiotherapists to remotely monitor patient progress. This work supports the notion that a smartphone-based solution could form the basis of a practical user application.

Spina et al. [21] also developed a smartphone-based rehabilitation system which used phone sensors to monitor exercises. Accelerometer and magnetometer data for upper body movements such as arm abductions and elbow circles was captured from a phone worn on a wrist holster. What was unique about this system was that it also provided a means of real-time audio feedback to users to help guide them through proper execution and form. Again, this work further demonstrates the feasibility of using a single smartphone as the sole sensing device for upper body rehabilitation exercise monitoring.

Chen et al. [9] also showed that wearable sensors could reliably measure shoulder range of motion, and, when paired with a smartphone app, it could also guide patients through home exercise programs. Their system used three sensors (each of which combined accelerometers and gyroscopes) worn on the sternum, upper arm, and wrist. These sensors communicated wirelessly with a mobile app to provide real-time movement feedback, track exercise completion, and allow remote physician supervision. Patients using the app showed higher exercise completion rates (around 80%) compared to those doing unsupervised home exercises (around 35%), suggesting that smartphone-integrated motion tracking can improve adherence to prescribed exercises.

3 What Was Done and How

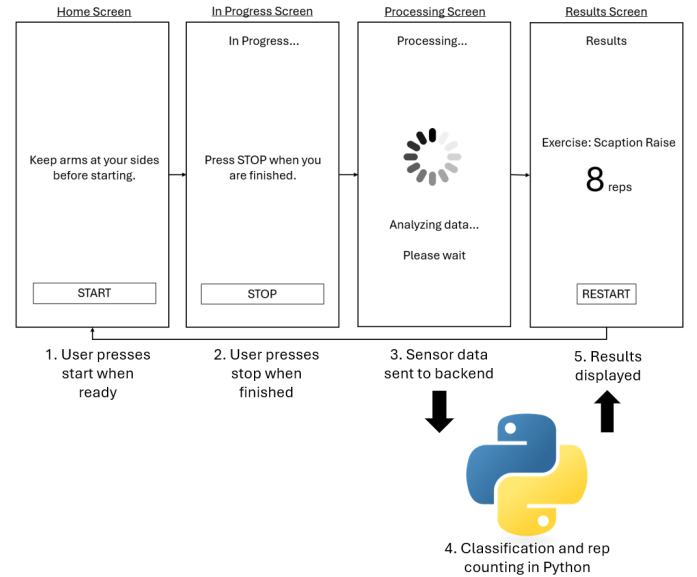


Figure 2: Diagram illustrating the application workflow: (1) the user presses start when ready, (2) the user presses stop when finished, (3) sensor data is sent to the backend, (4) the backend performs classification and repetition counting, and (5) results are displayed.

Our project consists of the following components:

- (1) Sensor data collected from four team members performing three exercises.

- (2) A peak counting algorithm for determining repetitions performed for three shoulder rehab exercises written in Python.
- (3) A machine learning model for classifying three shoulder rehab exercises also written in Python.
- (4) A backend server that hosts these Python scripts implemented with FastAPI and served with Uvicorn.
- (5) A React Native user interface consisting of four screens.

Figure 2 illustrates the overall application workflow, showing how user actions in the mobile app map to backend processing and final result display.

3.1 Data Collection

The raw sensor data is available in the `raw_data` folder of our GitHub repository. It was collected using the Phyphox mobile app and includes accelerometer and gyroscope readings. For the training set, all team members performed 15 repetitions of each exercise, while for the test set, each member performed a unique number of repetitions. Cleaned versions of the CSV files are provided in the `cleaned_data` folder.

During data collection, each team member wore a smartphone secured in an armband positioned above the right elbow. The device was oriented such that the screen faced away from the arm, and the top of the phone pointed toward the ground when the arm was in a resting position at one's side.

3.2 Repetition Counting

The peak counting algorithm (`peakCount.py` in our repo) determines the number of repetitions performed by processing sensor data from a phone's accelerometer and gyroscope. We use accelerometer data for counting Window Washing and Scaption Raise repetitions and gyroscope data for counting External Rotation repetitions. We transformed the sensor readings by computing the accelerometer's vector magnitude, isolating the gyroscope's y-axis, and inverting both signals so that troughs became peaks. We then smoothed the signals using a moving average to reduce noise and applied the SciPy function `scipy.signal.find_peaks` with minimum height and minimum peak distance parameters to get our repetition count.

3.3 Classification Model

Our classification logic can be found in the `classification.py` script in our repo. Here we extract features from accelerometer and gyroscope CSV files to train a model and then evaluate its accuracy. We first normalize the data using `np.linalg.norm` and then calculate its moving average to smooth the signal. Peaks are then detected using `scipy.signal.find_peaks` and features are extracted from these peaks. These features are the inputs for our classifier and consist of the ratio of gyroscope to accelerometer peak magnitudes, average peak values, and standard deviations.

The script loops through our training files (CSV files where each team member performed 15 repetitions of each exercise) and uses that data to train an `sklearn.ensemble.RandomForestClassifier`. The remaining CSV files are used as the test set. Their features are extracted, passed into the trained model to predict exercise types, and compared to the true labels to compute accuracy.

This approach to feature selection and the use of a Random Forest classifier was informed by guides on implementing and tuning Random Forest models in Python using Scikit-learn [12, 15].

3.4 Backend

The backend server for hosting these Python scripts was implemented using FastAPI and Uvicorn. FastAPI is a web framework for building APIs with Python [4] while Uvicorn is a lightweight web server that runs FastAPI applications [13]. Dependencies for these were installed with the command `pip install fastapi uvicorn` on our local machines.

The backend server logic can be found in the `main.py` file in our repo. At startup, this file loads the cleaned training data, extracts features, and trains the random forest classifier. It provides an HTTP POST endpoint (`/analyze`) which accepts JSON-formatted accelerometer and gyroscope readings from the frontend. Once this data is received, the server computes and returns three values:

- (1) The workout duration, computed from timestamp differences.
- (2) The repetition count, calculated using the logic in `peakCount.py`.
- (3) The predicted exercise type, determined using the classifier logic in `classification.py`.

3.5 User Interface

The user interface is made up of four screens written in React Native, an open-source mobile app framework that supports cross-platform development with a single codebase [14]. These four screens are shown in Figure 3. Each screen is defined in a different TypeScript file in our repo:

- (1) `index.tsx`: The home screen where the user presses a START button when they are ready to begin their exercise.
- (2) `in-progress.tsx`: The screen that handles the live workout. Using the `expo-sensors` library, the app subscribes to the phone's accelerometer and gyroscope (setting a sampling rate of 100 Hz). It waits three seconds before starting to capture sensor data, then, when the user has finished performing the exercise and presses the STOP button, the app saves the sensor data and navigates to the processing screen.
- (3) `processing.tsx`: An intermediate loading screen that appears while the collected data is sent to and processed by the backend.
- (4) `results.tsx`: A screen that displays the exercise identified by the classifier, the number of repetitions performed, and the total workout duration. It also provides users with the option to return to the Home Screen to restart.

Expo was used alongside React Native to reduce build and testing complexity [1]. Its `expo-sensors` API enabled access to the accelerometer and gyroscope for data collection, and the mobile app Expo Go was used to test the application directly on our mobile devices.

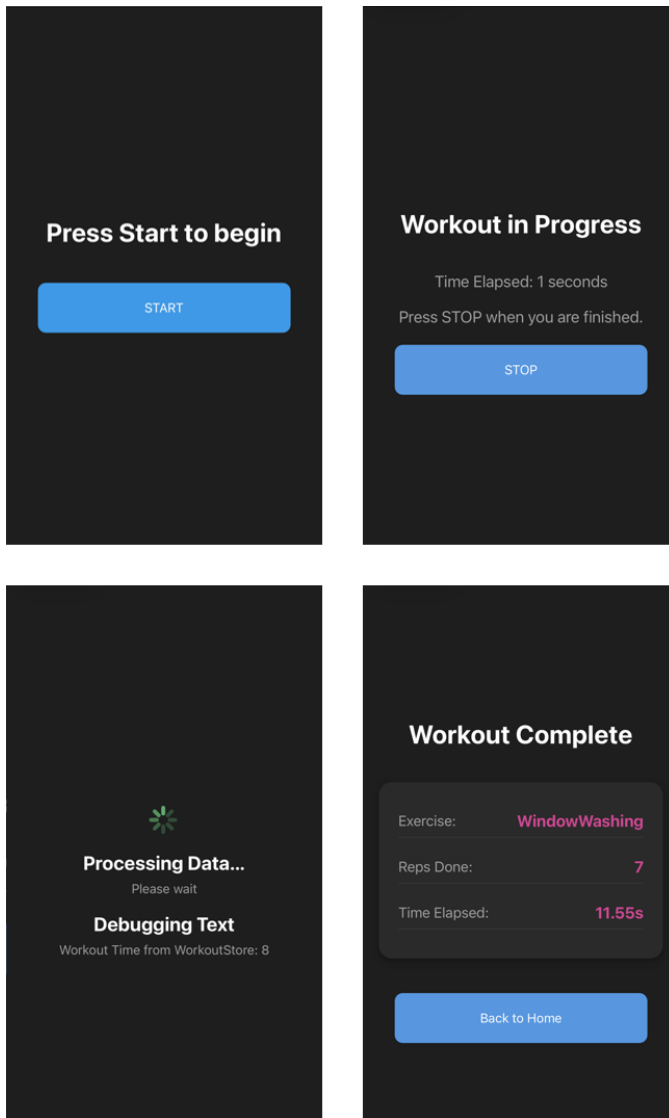


Figure 3: The four screens of the mobile application displayed in the order a user encounters them: Home screen, In-progress screen, Processing screen, and Results screen.

3.6 Linking Back to the Original Motivation

This project was originally motivated by how one of our team members has experienced recurrent shoulder issues requiring repeated physical therapy. Consistent and measurable rehabilitation can be a challenge for individuals outside of dedicated facilities. The work we’ve done in this project presents part of a potential solution that can provide users with feedback on their exercise performance. With further refinements, this could serve as an aid that complements therapy by allowing users to verify that they are executing movements correctly and by potentially maintaining a record of their activity.

4 Challenges & Alignment to Class

4.1 Challenges

As observed during our implementation, smartphone sensor accuracy is often inconsistent around different manufacturers and device models [16]. Each of the four team members involved utilized a different smartphone model for data collection. While our offline machine learning pipeline achieved a 91% classification accuracy using structured data recorded via Phypbox, we found that success levels varied between individual team members when this model was applied to live scenarios (i.e., through our phone app).

The primary technical challenge therefore involved the transition between the clean data environment, consisting of our offline test and training sets, to the noisy real life scenario of using our phone app. Although we achieved the stretch goal of developing a functional app, we found that the data produced by live user interactions demonstrated fluctuations in accuracy between different phone models.

4.2 Alignment to Class

Accomplishing this project involved navigating the Activity Recognition Chain (ARC) [8], from initial sensor data acquisition to classification and counting. By using the smartphone, which is arguably the most ubiquitous computing device in modern life, to classify and count exercises, we explored how invisible technology can be used to move the therapeutic experience from a clinical setting into the home. Because the transition to a live phone application introduced the aforementioned hardware inconsistencies, the development of the app itself (originally a stretch goal) highlighted the importance of the pre-processing and segmentation stages of the ARC and the ways in which live noisy data differs from offline or controlled test data.

5 Results

In order to understand how this application would be helpful, we interviewed seven participants in a think-aloud interview process where a team member would have the participant use the application as if they were exercising naturally. We had the user vocalize their thoughts in real-time and asked the user for feedback afterwards. We also had eleven potential users complete a 10-question survey to gain a deeper understanding on how users would respond to the application. We used these two different methods to evaluate the application because when we initially began this study, we determined that users found it important to notice growth while doing rehabilitation exercises. With that in mind, these evaluation formats provided us insights into that need both in practice and theoretically.

5.1 Qualitative Results From Interviews and Surveys

In the think-aloud interviews, we had the users vocalize their thought process and ask questions as they were interacting with the application and performing the exercises. The main goal of these interviews was to get a deeper understanding of the steps a user takes while using the app to ensure that it accurately counts

repetitions and classifies the exercise type.

Are users comfortable using the arm band?

Users did not like interacting with their phone while having the armband strapped to their bicep; they felt that it restricted their mobility and made it difficult to use the application in that position. Some users suggested a different placement, such as on the wrist via a smart watch instead of a phone. This suggestion is further supported by the fact that smart watches are already commonly used to track step count, heart rate, and other related metrics, making it a natural transition for users to also utilize them for exercise classification and repetition counting.

How accurately was the application able to classify the exercise and count repetitions?

The classification and repetition counting yielded a 75 percent accuracy rate. We believe discrepancies in these could be related to the type of phone the participant used the application on due to the sensors on iPhones and Androids collecting data in differing manners.

What are the pain points the users felt as they were interacting with the application?

Many users stated that the application should provide and receive auditory signals. An example of when an auditory signal could be used is during the count down from three to one when starting an exercise on the application (since we request users stand still for three seconds before beginning the exercise). This provides multiple cues across various modalities that helps ensure the user knows when the start the exercise. Users also commented that it would be helpful if the application could receive auditory signals from the user via spoken text into the mobile phone microphone (for example, saying “Start” and “Stop” when beginning and ending the exercise, respectively). This would further reduce friction of use by giving the user various options of how to interact with the application.

Furthermore, participants in the interviews mentioned it would be helpful to have a live auditory repetition counter so they can more precisely understand what counts as a repetition and what does not.

5.2 Quantitative Results From the Interviews and Surveys

With this survey, we attempted to to gain a deeper understanding of how users would perceive the application, what they would use it for, and key features they might find useful in future versions. We created the survey using a mix of multiple choice, 5 point Likert scale, and free response questions. (See appendix for a complete breakdown of the survey results).

Which feature of these features in an exercise app would be more valuable to you?

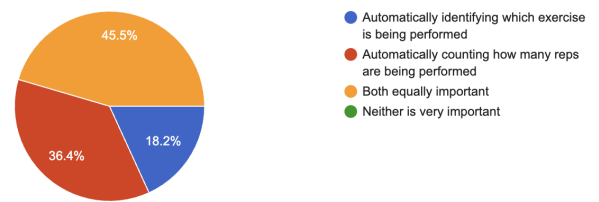


Figure 4: Survey results on exercise tracking feature preferences

Nine out of eleven survey participants indicated that they have performed some form of rehabilitative exercise/stretching and all the survey participants stated that they would benefit from some form of exercise tracking. Through this question, we aimed to gather a deeper understanding of what users would find most useful and the majority of our survey participants stated they would benefit from a way to have the app classify the exercise and count the repetitions for them.

How useful would it be for an app to automatically identify the exercise you are performing?

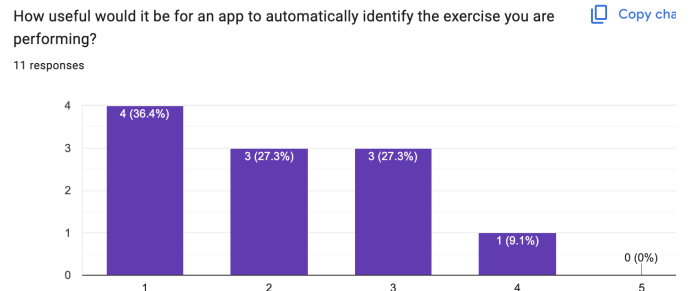


Figure 5: Survey results on exercise identification preferences. Please note 1 is most useful and 5 is least useful

When users were asked about the importance of having the application identify the exercise they are performing, 63 percent stated it would be helpful. Through the interviews, we learned that users would find it helpful to gauge if they are performing the exercise with correct form. If the application does not identify the activity correctly, there is a chance the sensors are picking up on a different kind of movement.

As a team, we understand that due to an insufficient sample size of participants, the results are statistically insignificant. We are hoping to investigate this further with more participants in the future.

6 Conclusion and Reflection

The technical goal of this project was to have an application which can perform shoulder exercise recognition using smartphone sensors, which was achieved. To achieve this, a complete machine learning pipeline was built with the stages of data collection with Phyphox, peak detection to determine repetitions, classification and evaluation. This resulted with 75% classification and repetition counting accuracy by participants in our interviews which shows that without any specialized medical devices, meaningful results can be accomplished.

Conducting the surveys and interviews provided insights on the usability of our application versus just technical success. 45% of participants considered automatic exercise identification and automatic repetition counting to be equally important, which matches our project scope. This type of feedback was only discoverable through direct user feedback.

Hardware variances proved to be a challenge as the sampling rates differences between iPhones and Pixel phones that were used for the initial data collection were significant. This represents a real limitation with our application – a tool such as this needs to accommodate the device that is being used for shoulder rehabilitation. In addition, the data collection and real-live data used different tools (Phyphox vs Expo Sensors) which added another layer of discrepancies as our model was trained using Phyphox.

Through our user research, the need to improve our design of the interaction was evident as the armband placement was more of a barrier than anticipated. Suggestions for this included changing usage to a smart watch as they already include exercise tracking functionality. Another usability improvement would be to add auditory cues to give the users indication when to start performing the exercises. After hitting start, there was countdown before starting but due to the placement of the phone, it was not visible to participants.

A takeaway for this project is the importance of user design in conjunction with technical design. Building the classifier was the main focus of the project, but the interaction design of our application was not iterated enough. Running small tests with users during development, whether it is surveys or interviews, would have helped to identify pitfalls with our approach and pivot as necessary. A takeaway from a technical perspective is the hardware differences between smartphones is directly linked to model performance as the data inconsistencies limited the classifiers performance.

6.1 Future Work

Naturally, future work includes addressing the device variances and developing a method that is device agnostic. Normalizing sampling rates across phones would enable better results in our classification and improve accuracy if done before training.

For the user design, future iterations should explore alternative device placements or using another smart device such as a smart watch that does not interfere with the movement of the arm. Also, the integration of auditory or haptic feedback as guide for users to use throughout the exercises. This quick-win would deliver a high-impact with our users and change the usability of our application.

Finally, more research is needed in shoulder rehabilitation exercises and sensor-detection; experimentation of more exercises would give insight on performance metrics variances between them. A larger participant pool with varying characteristics and demographics, would be an essential step for any clinical application.

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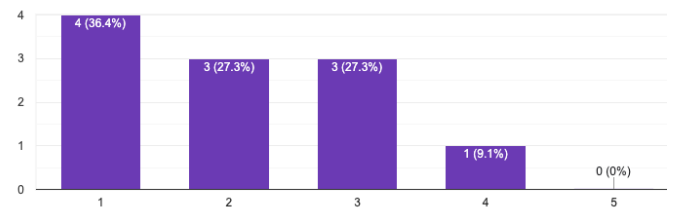
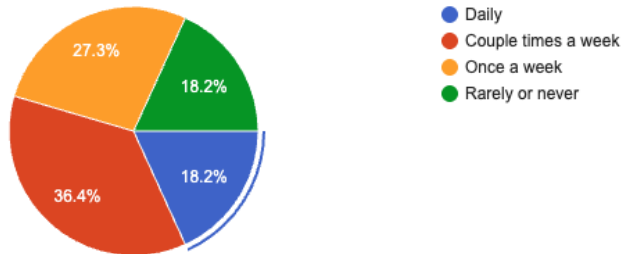
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7 Appendix

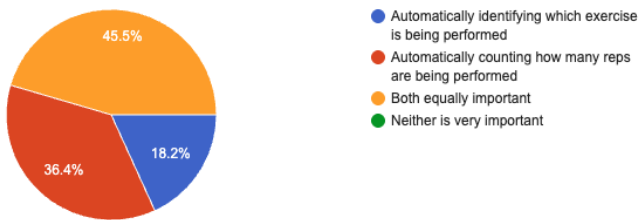
7.1 Survey Questions and Results

*Survey results are posted in the form of graphs.

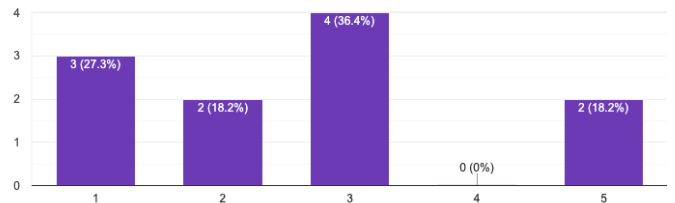
1 How often do you currently perform any kind of rehabilitative exercise such as physical therapy, stretching, injury recovery?



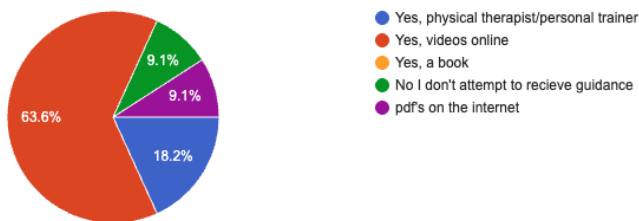
2 Which feature of these features in an exercise app would be more valuable to you?



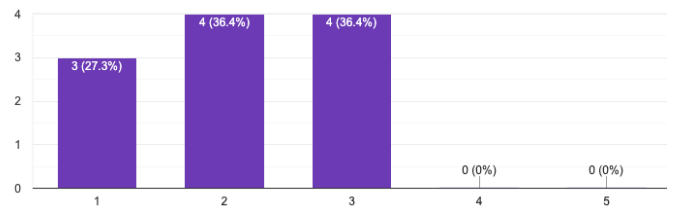
5 Rate how comfortable are you with using an armband to holster your phone to your arm while exercising?



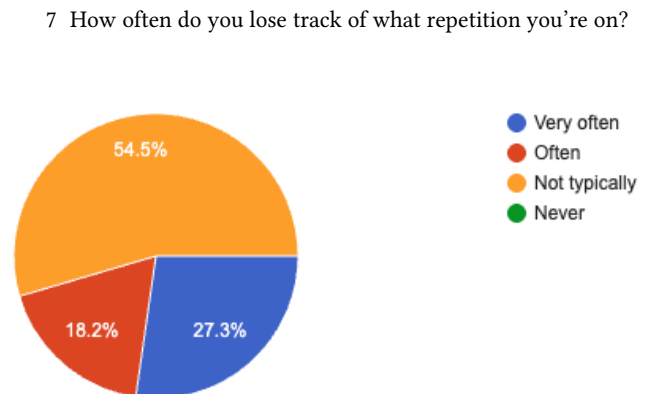
3 While performing these exercises do you try to receive guidance to ensure you are doing it properly? If so, how?



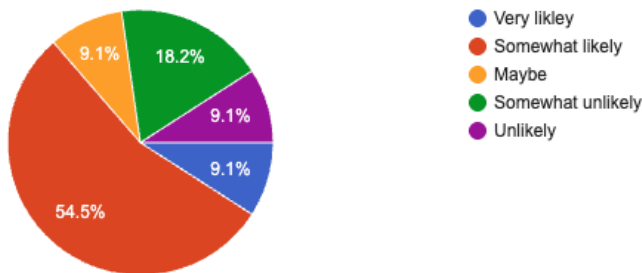
6 Rate how important real-time feedback is to you while exercising?



4 How useful would it be for an app to automatically identify the exercise you are performing?



- 8 If an app could count your reps and classify your exercise for you based on your movements, how likely are you to use it?



- 9 What use, if any, would you find in an application identifying the exercise and counting the repetitions for you? (responses listed below):

- Identifying exercise - helpful so I don't have to select from a list in the app; could save time counting reps - very useful as sometimes my mind wanders and I lose track
- Sometimes I forget to keep track of what I've done for workouts, so this could help automatically keeping track of what I did.
- could focus more on my form as opposed to counting a specific number of reps
- It will save time from having to manually input the workout and losing track of the reps.
- I would find it helpful to have all that information noted for me without having to write it down myself
- Improve accuracy and save time
- Somewhat useful
- I think it'd be nice for making sure I don't lose track of reps bc I lose count easily

- 10 What features would make a smartphone-based exercise tracking app most useful or trustworthy to you? (responses listed below):

- counting reps/tracking progress
- I would like to get personalized feedback based on the data from exercise sessions.
- real time feedback telling me whether or not I'm performing an exercise accurately or consistently in my form
- Trends over time, like how my body is progressing from my workouts
- Making sure my information is safe, to the point where maybe I don't even need to link anything to my email but the user account is just fully separate
- Reliable exercise recognition and rep counting that works across different angles and body types. Integrates with wearables and/or health apps.

- Reliable heart rate monitor, and identifiable exercise tracking
- accurate rolling count of my reps and saved in app

7.2 Think Aloud Interview Feedback

1 Participant 1

- Wish it would accurately count the reps and classify the exercise every time
- Too awkward pressing start and stop on your upper arm, some external start button would be nice
- Have it work on for exercises besides shoulder exercises
- Have it ding after waiting 3 seconds
- Have it automatically detect when you start doing other exercises

2 Participant 2

- Can't see the phone screen, could it be rotated to the front of the arm instead?
- Would like a confirmation noise for each rep being performed
- Pushing the button on the phone is too hard
- Waiting three seconds... I'd like some confirmation noise to let me know when to start
- Make it give a warning if you're not standing still for the 3 seconds it expects you to

3 Participant 3

- Takes a bit too long to process data
- Auditory signals that what you're doing in being recorded
- Example of what the exercise should look like
- Maybe a list of what kinds of exercises can be done
- Kinda feels uncomfortable to have it on the arm
- Not enough instructions on what I can do with this app
- The about section could have information on what exercises I can do and how to do them
- When the phone is stationary it automatically says window washing and gives a random number of reps
- "Debugging text" on the wait screen shouldn't be user facing
- The time elapsed should only show the time you were actually working out and not include the 3 seconds of wait time

4 Participant 4

- Should tell you to put your hands by your side first before starting a countdown
- Have some vibration to know when to start, because you cannot see the app when arms are by your side unless you're waltz chamberlain
- vibrating on each rep, when the app senses you did the exercise properly to let you know it's being counted,
- Spun the arm strap a bit to see the app better, not easy to read
- Said 12 reps instead of 7 done

5 Participant 5

- Unclear of what exercises are supported by the app

- Didn't know when to begin the exercise after pressing start and putting hands to side
- Awkward to have to end the recording while the device is still on the arm
- No way to export the results
- Would like a way to view historical data on exercises/rep counts, rather than the results clearing every time

6 Participant 6

- The start position needs to be clearer, like waiting for the user to be still before starting the countdown
- Phone placement is too restrictive (should allow both arms, phone facing up/down etc)
- It should have some feedback about what exercise/rep you're currently doing (live rep count)
- Hard to press stop, etc.

7 Participant 7

- Might be better as an application for a smart watch so you can interact with it better

- Have a description of what the exercises are
- Tell you what rep you're on or select how many reps you wanna do and gives an auditory signal when you have reached the right number of reps
- Have a video of how to do the exercises
- Or tells you haven't reached the right number of reps if you stopped early
- Selection of all the exercises you can do
- Auditory signal of when the wait three seconds time is done
- Would be nice if it accepted auditory cues – such as “start” or “stop” and not have to interact with the screen at all
- Keep track of how many reps you do over time to monitor progress

7.3 GitHub Repository

The code for this project can be found here: <https://github.com/gatech-edu/vwhite36/MUCProject>